



Effects of Mn promotion on the activity and selectivity of Co/SiO₂ for Fischer–Tropsch Synthesis

Arne Dinse^{a,1}, Max Aigner^b, Markus Ulbrich^b, Gregory R. Johnson^a, Alexis T. Bell^{a,*}

^a Department of Chemical and Biomolecular Engineering, University of California, Berkeley, CA 94720, USA

^b Department of Chemistry, TU München, D-85748 Garching, Germany

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ABSTRACT

An investigation has been carried out of the effects of Mn promotion on the activity and product selectivity of Co/SiO₂ for Fischer–Tropsch Synthesis (FTS). All experiments were conducted with an H₂/CO feed ratio of 2.0 at either 1 atm or 10 atm and a temperature of 493 K. At 1 atm, the rate of CO consumption decreased with CO conversion, whereas the selectivities to all products were unaffected. However, the olefin to paraffin (O/P) ratio of the C₂–C₄ fraction decreased with increasing CO conversion due to increased hydrogenation of olefins with increasing space time. Mn promotion increased the rate of CO consumption at Mn/Co ratios of 0.05, decreased the selectivity to methane, and increased the selectivity to C₅+ products, but had no effect on the selectivity to C₂–C₄ products. Raising the pressure to 10 atm increased the CO consumption rates for both unpromoted and Mn-promoted Co/SiO₂, but the rate of reaction was lower for the Mn-promoted catalyst due to a higher level of CO inhibition. CO conversion at 10 atm had no effect on the rate of CO consumption for Co/SiO₂ and caused a slight increase in the rate for Mn-promoted Co/SiO₂. The intrinsic O/P ratio was higher at 10 atm than at 1 atm. In contrast to what was observed at 1 atm, at 10 atm, C₂–C₄ of the C₂–C₄ fraction incorporated into growing hydrocarbon chain, leading to a decrease in C₁–C₄ selectivities and an increase in C₅+ selectivities with increasing CO conversion. The observed effects of Mn promotion on catalyst activity and product selectivity are discussed in terms of the mechanisms for CO hydrogenation and hydrocarbon chain growth.

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1. Introduction

Fischer–Tropsch Synthesis (FTS) has experienced a resurgence of interest in recent years because it offers a means for converting synthesis gas derived from non-petroleum feedstocks, such as natural gas, coal, and biomass, to transportation fuels [1–4]. The principal hydrocarbon products of FTS are methane (C₁), petroleum gas (C₂–C₄), gasoline (C₅–C₁₁), diesel (C₁₂–C₂₀), and wax (C₂₁+), which can ultimately be cracked into additional gasoline and diesel. The formation of large amounts of methane is undesirable, since it represents a loss of carbon that otherwise could be directed toward liquid products. The C₂–C₄ fraction is too light to be used for liquid transportation fuels; however, the alkenes contained in this fraction can be converted to higher molecular weight products by metathesis or oligomerization [5].

While most of the original FTS processes used Fe-based catalysts, the interest today has shifted toward Co-based catalysts, since they exhibit better selectivity to diesel and wax [6]. For this reason, a large fraction of recent work on FTS catalysts has focused

on increasing the activity of Co catalysts and altering the product distribution in order to minimize the formation of C₁ and maximize the formation of C₅+ products. It has generally been observed that operating at lower temperatures and higher reaction pressures favors the formation of C₅+ products and disfavors the formation of C₁ [6–16]. There is also evidence that similar effects can be achieved by operating at higher synthesis gas conversions due to the reincorporation of alkenes into the chain growth process [3,17–19].

While support composition has a minor [20] effect on the product distribution and reaction rates [11,21], modification of catalysts with metal oxide promoters can alter reaction rates, product distribution, and catalyst stability [21–30]. A promoter of particular interest is Mn [31–38]. In a series of studies conducted at 1 atm on Co/TiO₂, Weckhuysen and coworkers [20,39–43] have reported that Mn promotion increases catalyst activity, decreases C₁ formation, increases the olefin to paraffin (O/P) ratio of the C₂–C₄ fraction, and enhances the formation of C₅+ products. Only limited observations of the effects of Mn promotion have been reported for FTS carried out at high pressures. For example, Das and coworkers reported that at 10 atm and 493 K, Mn promotion increases the reaction rate and selectivities to C₁ and C₂–C₄ products and decreases the C₅+ selectivity and O/P ratio [44]. What has not

* Corresponding author. Fax: +1 510 642 4778.

E-mail address: bell@cchem.berkeley.edu (A.T. Bell).

¹ Current address: BP Conversion Technology Center, Naperville, IL 60563, USA.

been addressed by these studies, though, are the systematic effects of reaction conditions—temperature, pressure, and CO conversion—and the extent of Mn promotion on the activity and product selectivity of Mn-promoted Co for FTS. Here, we present a comprehensive study aimed at investigating the influence of reaction conditions and Mn promotion on these metrics. Experiments were conducted at pressures of 1 and 10 atm and CO conversions of 2–44% using SiO₂-supported Co both unpromoted and promoted to varying degrees with Mn.

2. Experimental

2.1. Catalyst preparation

Silica-supported Co (1 g) was prepared via incipient wetness impregnation of SiO₂ (PQ corporation, pore diameter 27 nm) using an aqueous solution (2.4 ml) of 0.67 g Co(NO₃)₂·6H₂O (Sigma–Aldrich) and varying amounts (0–0.28 g) of Mn(COOCH₃)₂·4H₂O (Aldrich) under constant stirring for 90 min followed by drying for 18 h at room atmosphere. Co weight loading was held constant at 12% with respect to the sum of Co metal and SiO₂ support, and Mn was added to achieve molar Mn/Co ratios of 0, 0.05, 0.125, 0.25, and 0.5. Metal weight loadings were confirmed using ICP (Galbraith Laboratories). Each of the as-prepared catalysts was placed in a quartz reactor and then reduced in H₂ (Praxair, 99.999% purity) flowing at 100 cm³/min as temperature was ramped at 4 K/min from 298 to 673 K and then held for 120 min at 673 K. The reduced catalyst was cooled to room temperature at a rate of 10 K/min and passivated in 0.5% O₂ in He (Praxair 99.999% purity) at a flow rate of 100 cm³/min for 90 min before storage.

2.2. Static H₂-chemisorption

Static chemisorption of H₂ was used to determine the surface concentration of reduced Co. Experimental runs were carried out in a Micromeritics ASAP 2020 instrument. Samples of 0.25–0.75 g were heated to 673 K under H₂ atmosphere at 4 K/min and held at this temperature for 120 min. Following reduction, samples were evacuated to 5×10^{-5} Torr at 653 K and cooled to 373 K at 10 K/min. Subsequently, adsorption and desorption isotherms were measured. Dispersion of unpromoted Co/SiO₂ was calculated based on the equation as reported by Reuel and Bartholomew assuming spherical metal crystallites of uniform diameter with a site density of 14.6 atoms/nm² and that H₂-chemisorption takes place at a ratio of H/Co of 1:1 [45].

2.3. Oxygen titration

Oxygen titration of reduced catalyst samples was used to determine the fraction of the Co in each sample that had been reduced. For each run, 250–500 mg of catalyst was loaded into a quartz reactor. The catalyst was then reduced in flowing H₂ 100 cm³/min while the temperature was raised at 4 K/min from 298 K to 673 K and then held at this temperature for 120 min. The reactor effluent was monitored at all times by online mass spectrometry (MKS Minilab). To remove any physisorbed H₂ from the catalyst surface, the system was flushed for 30 min with He flowing at 100 cm³/min until no more H₂ was detected. Subsequently, oxygen titration measurements were initiated. Pulses of O₂ were supplied from a sample loop (0.421 cm³) connected to a 4-way switching valve (Swagelok 40 series). Pulse sizes of 18 μmol were created using 100% O₂ (Praxair, 99.999%). O₂ not absorbed by the sample was detected using online mass spectrometry. Each pulse was injected after the previous signal had decreased back to the baseline

value, and the number of completely and partially absorbed oxygen pulses was quantified in order to calculate the total amount of O₂ absorbed.

2.4. Catalytic testing

Catalyst performance was measured at 1 atm using a quartz glass reactor and at 10 atm using a 316 stainless steel (SS) reactor with an inner diameter of 10 mm. For the experiments conducted at 10 atm, the reactor pressure was maintained using a backpressure regulator (Praxair, spring loaded). The background activity of both types of reactors filled with silica was found to be negligible up to 673 K. All measurements of catalyst activity and selectivity were carried out with 100 mg of catalyst diluted with silica (weight ratio 1–1). Prior to exposure to FTS reaction conditions, the catalyst sample was reduced in H₂ (Praxair, 99.999%) flowing at 100 cm³/min H₂ as the temperature was ramped at 4 K/min from 298 K to 673 K, held at this temperature for 120 min, and then cooled to the reaction temperature (493 K) at 10 K/min. The catalyst was then aged in a H₂/CO/Ar mixture (62/31/7, Praxair 99.5%, aluminum cylinder) flowing at 100 cm³/min for 4 h at 1 atm and for 20 h at 10 atm, respectively, after which time CO conversion and product selectivities remained constant. All experiments were carried out in the absence of mass transfer limitations determined by satisfaction of the Weisz–Prater criterion [46]. Temperature measurements were made with a thermocouple positioned in the center of the catalyst bed. Several experiments were made to identify the extent of temperature gradients in the bed. These were found to be less than 6 K and had very little effect on either the activity or the distribution of final products. The CO conversion was altered by changing the reactant (H₂/CO = 2) flow rate from 5 cm³/min to 120 cm³/min.

Exploratory experiments revealed that the optimal temperature for forming C₅₊ products at a high rate was 493 K. Consequently, all further investigations of the effects of Mn promotion, CO conversion, and pressure were carried out at this temperature.

The reactor effluent was analyzed online using a gas chromatograph (GC, Agilent 6890). A capillary column (HP-1) connected to a flame ionization detector (FID) was used to analyze hydrocarbon products. CO consumption was determined using a packed column (Porapak Q 2701PC, Alltech) connected to a thermal conductivity detector (TCD) using Ar as an internal standard. The oven temperature was held at 213 K for 5 min and then ramped up to 513 K at 50 K/min, after which it was maintained at 573 K for 5 min. GC inlets and detectors were maintained at 523 K. Calculation of gas-phase concentrations was done using measured response factors. Calibrations for C₁ to C₄ hydrocarbons were made using gas-phase concentrations in the target range of product concentrations and linear scaling as a function of carbon atoms for higher carbon numbers. All C₅₊ products were analyzed using the sum of all isomers detected with respect to the specific carbon number.

Carbon conversion, $X_{\text{CO}}^{\text{CO}}$, was determined from the difference in CO concentrations measured at the inlet and outlet of the reactor using Ar as an internal standard according to the following equation:

$$X_{\text{CO}}^{\text{CO}} = \frac{\frac{C_{\text{CO}}^0}{C_{\text{Ar}}^0} - \frac{C_{\text{CO}}}{C_{\text{Ar}}}}{\frac{C_{\text{CO}}^0}{C_{\text{Ar}}^0}} \quad (1)$$

Here, $\frac{C_{\text{CO}}}{C_{\text{Ar}}}$ is the ratio of the gas-phase concentrations of CO to Ar measured at the reactor bypass before and after experimental runs and $\frac{C_{\text{CO}}^0}{C_{\text{Ar}}^0}$ the ratio of gas-phase concentrations of CO to Ar during reaction. This method of determining CO conversion could be used reliably for CO conversions greater than 10%.

At 1 atm, CO conversion, $X_{\text{CO}}^{\text{HC}}$, was also calculated from the sum of all products detected in the reactor effluent using the following equation:

$$X_{\text{CO}}^{\text{HC}} = \frac{Q_{\text{out}} \sum_{i=1}^n v_i C_i}{Q_{\text{in}} C_{\text{CO}}^0} \quad (2)$$

In this expression, Q_{out} is the volumetric flow at the reactor outlet during reaction, Q_{in} is the volumetric flow at the reactor inlet during reaction, v_i is the stoichiometric factor, C_i is the concentration of hydrocarbon with carbon number i , and C_{CO}^0 is the concentration of CO at the reactor inlet calculated based on the syngas composition of the feed cylinder. The deviation of the carbon balance calculated from CO consumption and the sum of all products, reflected by the ratio of Eqs. (1) and (2), was equal to or smaller than $\pm 10\%$ at all reaction conditions, and CO conversions higher than 10% due to the standard deviation of CO/Ar ratios (3.95 ± 0.05) were determined via GC prior to reaction.

At 10 atm, carbon selectivities were calculated using Eqs. (3) and (4): C_{5+} selectivities were calculated from the difference between the overall conversion of CO determined from Eq. (1) and the conversion of CO to C_1 to C_4 products, all of which are in the gas phase. This approach facilitates product analysis because high molecular weight products are formed and wax collection including offline wax analysis requires several days. Carbon balances were within $\sim 10\%$ maximum deviation as determined by comparing online analysis of gaseous products and offline GC analysis of condensed wax products.

$$S_{C_1-C_4} (\%) = \frac{Q_{\text{out}} \sum_{i=1}^4 v_i C_i \cdot 100}{Q_{\text{in}} C_{\text{CO}}^0 X_{\text{CO}}^{\text{CO}} (\%)} \cdot 100\% \quad (3)$$

$$S_{C_{5+}} (\%) = \frac{Q_{\text{in}} C_{\text{CO}}^0 X_{\text{CO}}^{\text{CO}} (\%) - Q_{\text{out}} \sum_{i=1}^4 v_i C_i \cdot 100}{Q_{\text{in}} C_{\text{CO}}^0 X_{\text{CO}}^{\text{CO}} (\%)} \cdot 100\% \quad (4)$$

CO consumption rates were calculated using Eq. (5).

$$r_{\text{CO}} = \frac{F_{\text{CO}}^0 X_{\text{CO}}^{\text{CO}}}{W_{\text{Co}}} \quad (5)$$

In this expression, F_{CO}^0 is the molar flow rate of CO at the reactor inlet and W_{Co} is the weight of Co metal within the catalyst bed.

3. Results

3.1. Effects of Mn promotion on catalyst composition, structure, and chemisorption characteristics

Table 1 lists the average size of the Co particles calculated from the peak width at half maximum of XRD features using the Scherrer equation. XRD features were subject to baseline correction,

stripping of K_{α} , and correction for instrumental line broadening. Prior to determining the XRD pattern, samples were fully oxidized in O_2 at 673 K for 30 min (see Experimental). The observed XRD pattern showed features only for Co_3O_4 and none for the presence of CoO or Co(0). Co particle diameters were calculated assuming the partial reduction of Co_3O_4 to Co and taking into account the differences in the densities of Co and Co_3O_4 . The average Co particle size for unpromoted Co/SiO₂ was 6.3 nm, in good agreement with the value calculated from H₂-chemisorption experiments (7.5 nm). As the Mn/Co (0) ratio was increased from 0 to 0.5, the average Co particle size decreased monotonically down to 3.9 nm.

The uptakes of H₂ on Co/SiO₂ as a function of the level of Mn promotion are presented in Table 1. The H₂ uptake per gram catalyst initially decreased strongly upon promotion with small amounts of Mn (Mn/Co = 0.05), remained constant in the range of Mn/Co = 0.05–0.25, and then decreased again strongly at Mn/Co = 0.5. The dispersion and average particle size of unpromoted Co/SiO₂ were determined to be 15% and 7.5 nm, respectively, from the H₂ uptake for this sample. No uptake was found for Mn supported on SiO₂.

The results of O₂ pulsing experiments, undertaken to determine the percentage of Co sites that had been reduced from Co_3O_4 to Co, are also shown in Table 1. For Mn/Co = 0, 83% of all Co sites originally present as Co_3O_4 were reduced to metallic Co. With increasing Mn/Co ratio, the reducibility of Co_3O_4 decreased rapidly and then leveled off for Mn/Co = 0.5, for which 68% of all Co originally present as Co_3O_4 was reduced to metallic Co.

3.2. Effects of Mn promotion on activity and product distribution at 1 atm

Kinetics data were collected at 1 atm varying Mn/Co ratio and feed flow rate in order to determine the effects of these variables on the rate of CO consumption and the distribution of products formed. Fig. 1 (left) shows the influence of Mn/Co ratio on the rate of CO consumption for data taken at 493 K and extrapolated to zero residence time, as well as data acquired at CO conversions of 15% and 25%. The rate of CO consumption extrapolated to 0% CO conversion rose rapidly with Mn/Co ratio, reached a maximum at Mn/Co = 0.05, and then declined with further increase in the Mn/Co ratio. The maximum catalyst activity for Mn/Co = 0.05 was about 50% higher than that for Mn/Co = 0. With increasing extent of CO conversion, the maximum in CO consumption rate at Mn/Co = 0.05 decreased strongly. Fig. 1 (right) shows that CO consumption rates strongly decreased with increasing CO conversion at 1 atm. The observed decrease was between 33% and 56% in a CO conversion range of 2–40%. It is notable that the influence of Mn/Co ratio on the rate of CO consumption seen in Fig. 1 is quite similar to that recently reported by den Breejen et al. for FTS

Table 1

H₂ uptakes of Co/SiO₂ with varying ratios of Mn/Co measured with TPD. Samples were reduced at 673 K for 2 h, cooled to room temperature under H₂, and purged with He before measurements were initiated. O₂ pulse oxidation with 10% O₂ in He after reduction of Co/SiO₂ and Mn-promoted Co/SiO₂ for 2 h in H₂. Temperature = 673 K, pressure = 1 atm, total flow = 100 ml/min. XRD with samples oxidized in air for 2 h. Calculation of Co particle size using the Scherrer equation based on Co_3O_4 features and ratio of Co_3O_4 and Co densities.

Sample	H ₂ uptake (mmol g ⁻¹)	O ₂ uptake (mmol g ⁻¹)	Co(0) ³ (%)	d(Co) ² (nm)
Mn/Co = 0	0.11 ¹	1.1	83	6.3
Mn/Co = 0.05	0.04	0.96	71	5.5
Mn/Co = 0.125	0.06	0.92	68	4.7
Mn/Co = 0.25	0.045	1.01	74	4.4
Mn/Co = 0.5	0.005	0.9	66	3.9
Mn/SiO ₂	0	0	0	–

¹ Dispersion Mn/Co = 0: 15% (7.5 nm) based on [39].

² Diameter of Co(0) particles from XRD.

³ Based on oxidation of Co(0) to Co_3O_4 .

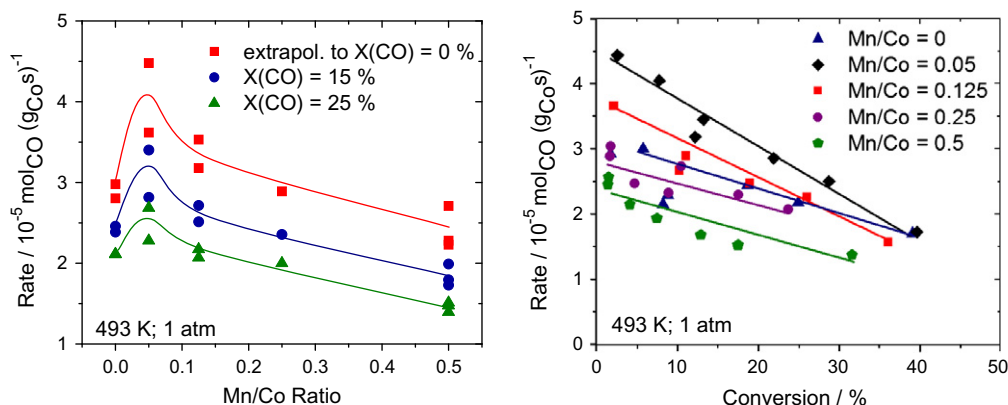


Fig. 1. Comparison of CO consumption rates at different Mn/Co ratios including reproducibility measurements (left) as well as rates as a function of CO conversion at 493 K and 1 atm. Feed ratio $\text{H}_2/\text{CO}/\text{Ar} = 2/1/0.2$, flows = 2.5–100 ml/min, catalyst mass = 100 mg.

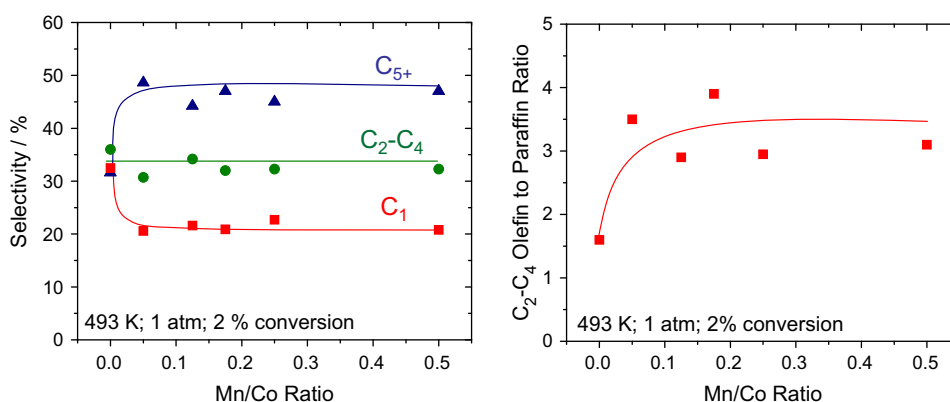


Fig. 2. C_1 , C_2 – C_4 , and C_{5+} selectivities (left) and O/P ratios within the C_2 to C_4 product fraction (right) as a function of Mn/Co ratio at 493 K and 1 atm. Feed ratio $\text{H}_2/\text{CO}/\text{Ar} = 2/1/0.2$, flows = 2.5–100 ml/min, catalyst mass = 100 mg, CO conversion = 2%.

carried out using a Mn promotion of $\text{Co}/\text{Pt}/\text{SiO}_2$ under conditions identical to those used in the present study [31].

As seen in Fig. 2 (left), the selectivity to C_1 was highest for unpromoted Co/SiO_2 and decreased strongly upon promotion with small amounts of Mn ($\text{Mn}/\text{Co} = 0.05$), beyond which the selectivity was constant for Mn/Co ratios up to 0.5. The decrease in the selectivity to C_1 was accompanied by an increase in the C_{5+} selectivity, whereas the C_2 – C_4 selectivity was completely independent of the Mn/Co ratio. Fig. 2 (right) and Table 2 show that the Mn/Co ratio did, however, affect the O/P ratio of the C_2 – C_4 fraction and the O/P ratio at a given carbon number. The O/P ratio of the C_2 – C_4 fraction increased by over a factor of two when the Mn/Co ratio was raised from 0 to 0.05, but then remained constant for further increases in the Mn/Co ratio. Table 2 reveals that the effect of Mn/Co ratio on individual components was not equivalent and was not monotonic. It is noted that the O/P ratios for the C_3 and C_4 fractions were an order of magnitude higher than that for the C_2 fraction. All further experiments were conducted with Co/SiO_2 promoted with Mn at the level of $\text{Mn}/\text{Co} = 0.125$ and with unpromoted Co/SiO_2 for reference.

Table 3 compares the activities of these two catalysts at 15% CO conversion. While the rates of formation of C_1 , C_2 – C_4 , and C_{5+} hydrocarbons were nearly equivalent for unpromoted Co/SiO_2 , the rates of formation of these products changed to differing degrees upon promotion with Mn. Most notably, the formation of C_1 dropped from 32% to 21% of the total CO consumed, whereas the formation of C_2 – C_4 hydrocarbons remained at 33% of the CO consumed, and the formation of C_{5+} hydrocarbons increased from 32% to 48% of the CO consumed. Fig. 3 shows that the selectivities to these three classes of products remained nearly constant as the conversion of CO increased up to 25%. The most notable change with CO conversion was in the distribution between olefins and paraffins in the C_2 – C_4 fraction, as seen in Fig. 3. The trends observed for the C_2 – C_4 fraction were similar to those for the individual C_2 , C_3 , and C_4 products (see Supplementary material Figure S1). For both unpromoted and promoted Co/SiO_2 , the selectivity to olefins decreased and the selectivity to paraffins increased with increasing CO conversion as seen in Fig. 4. This effect, though, was much stronger for unpromoted than

Table 2

Olefin to paraffin ratios for carbon numbers C_2 , C_3 , and C_4 at different Mn/Co ratios determined at a CO conversion of 2%.

	Mn/Co = 0	Mn/Co = 0.05	Mn/Co = 0.125	Mn/Co = 0.175	Mn/Co = 0.25	Mn/Co = 0.5
C_2	0.3	0.7	0.5	0.7	0.6	0.6
C_3	2.7	7.7	5.2	7.6	5.4	5.6
C_4	2.3	5.0	3.6	6.4	3.6	3.9

Table 3
Comparison of CO consumption and hydrocarbon formation rates for unpromoted Co/SiO₂ and Mn-promoted Co/SiO₂ with a Mn/Co ratio of 0.125. Pressure = 1 atm, temperature = 493 K, 15% conversion. Calculations based on selectivities given in Fig. 3.

Catalyst	CO (10^{-5} mol/g _{Co} /s)	C ₁ (10^{-5} mol/g _{Co} /s)	C ₂ –C ₄ (10^{-5} mol/g _{Co} /s)	C ₅ –C ₁₁ (10^{-5} mol/g _{Co} /s)
Mn/Co = 0	2.3	0.74	0.76	0.74
Mn/Co = 0.125	2.4	0.50	0.79	1.15

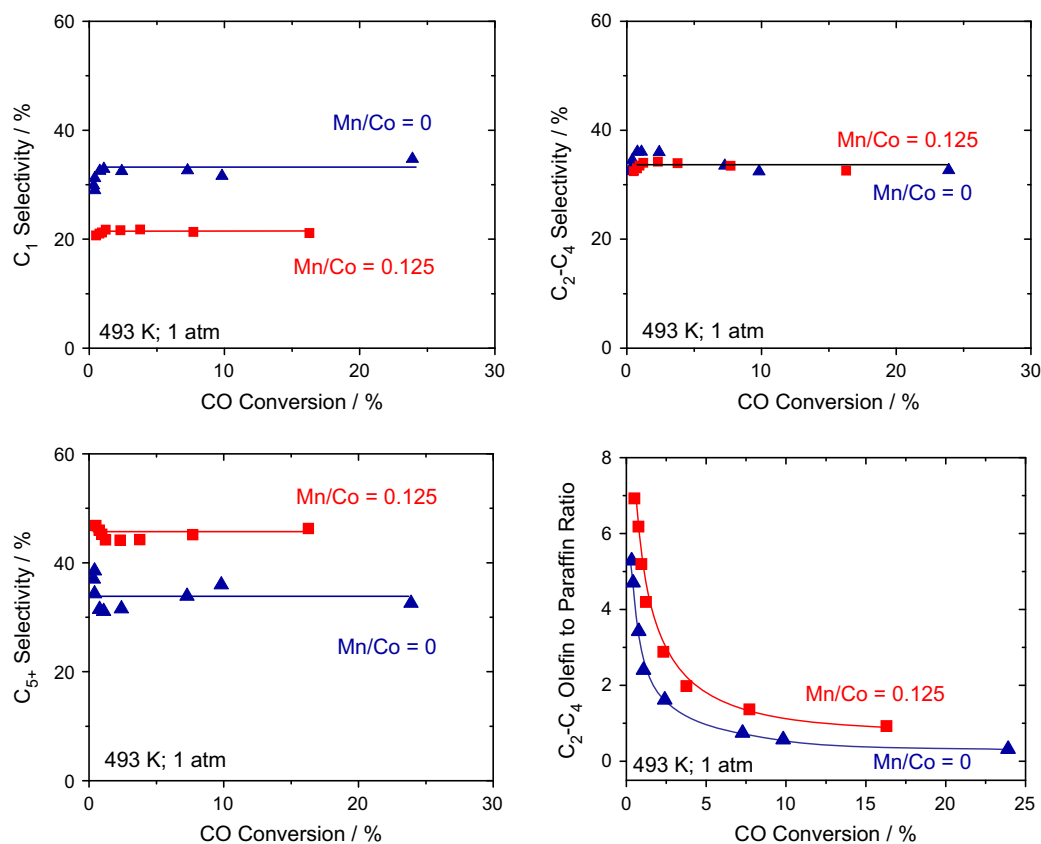


Fig. 3. Effect of CO conversion on selectivities to C₁ (top left), C₂–C₄ (top right) and C₅+, fraction (bottom left) and O/P ratio within the C₂–C₄ fraction (bottom right) at 493 K and 1 atm for catalyst with Mn/Co ratios of 0 and 0.125. Feed ratio H₂/CO/Ar = 2/1/0.2, flows = 2.5–100 ml/min, catalyst mass = 100 mg.

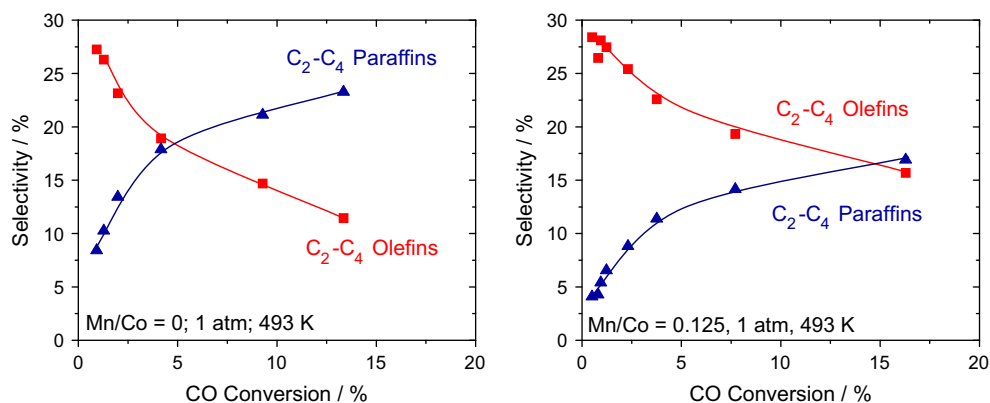


Fig. 4. C₂–C₄ olefin and paraffin selectivity as a function of CO conversion at 473 K and 1 atm for catalysts without Mn promotion (left) and Mn-promoted Co/SiO₂ with a Mn/Co ratio of 0.125 (right). Feed ratio H₂/CO/Ar = 2/1/0.2, flows = 2.5–100 ml/min, catalyst mass = 100 mg.

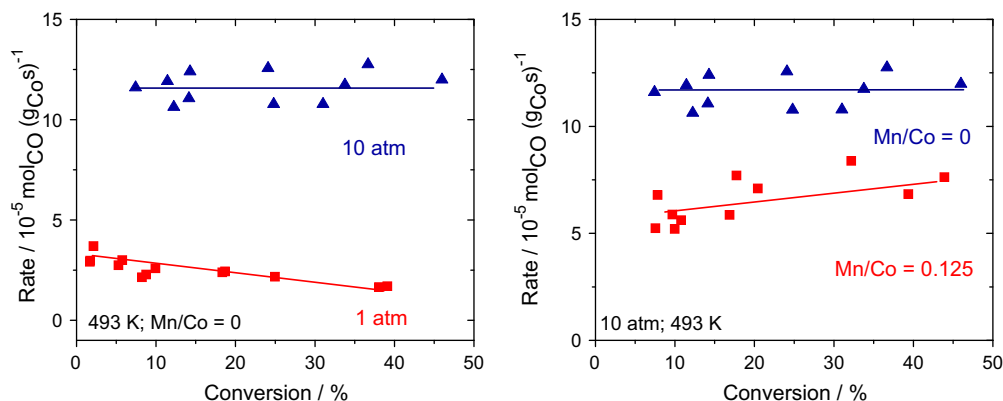


Fig. 5. Effect of increasing pressure on CO consumption rates for unpromoted Co/SiO₂ (left) and comparison of unpromoted and Mn-promoted Co/SiO₂ at 10 atm (right). Temperature = 493 K, feed ratio H₂/CO/Ar = 2/1/0.2, flows = 20–120 cm³/min, catalyst mass = 100 mg.

Mn-promoted Co/SiO₂. The effects of CO conversion and Mn promotion on C₂, C₃, and C₄ products showed trends that were similar to those seen in Fig. 3 (Figures S2 and S3).

3.3. Effect of Mn promotion on activities and product distribution at high pressures

Fig. 5 (left) shows that upon increasing the total pressure from 1 atm to 10 atm, the CO consumption rate of unpromoted Co/SiO₂ increased by 4.8-fold at 10% CO conversion and by 7.2-fold at 40% CO conversion. In contrast to what was observed at 1 atm, the rate of CO consumption for Co/SiO₂ was constant with increasing CO conversion at 10 atm. Fig. 6 shows that in contrast to what was

observed at 1 atm, the C₁ and C₂–C₄ selectivities decreased, the C₅+ selectivity increased substantially for unpromoted Co/SiO₂, and the O/P ratio of the C₂–C₄ fraction decreased slightly.

For Mn-promoted Co/SiO₂, the increase in reaction rates was less pronounced when the pressure was raised from 1 to 10 atm, as can be seen from Fig. 5 (right). Mn promotion caused a 2.3-fold increase at 10% CO conversion and a 3.2-fold increase at 40% CO conversion. Mn promotion of Co/SiO₂ also resulted in an increase in the rate of CO consumption with increasing CO conversion at 10 atm. Fig. 7 shows that Mn promotion of Co/SiO₂ substantially decreased the selectivity to C₁ products and increased the selectivity to C₂–C₄ products and C₅+ products. In addition, the O/P ratio of the C₂–C₄ fraction strongly increased.

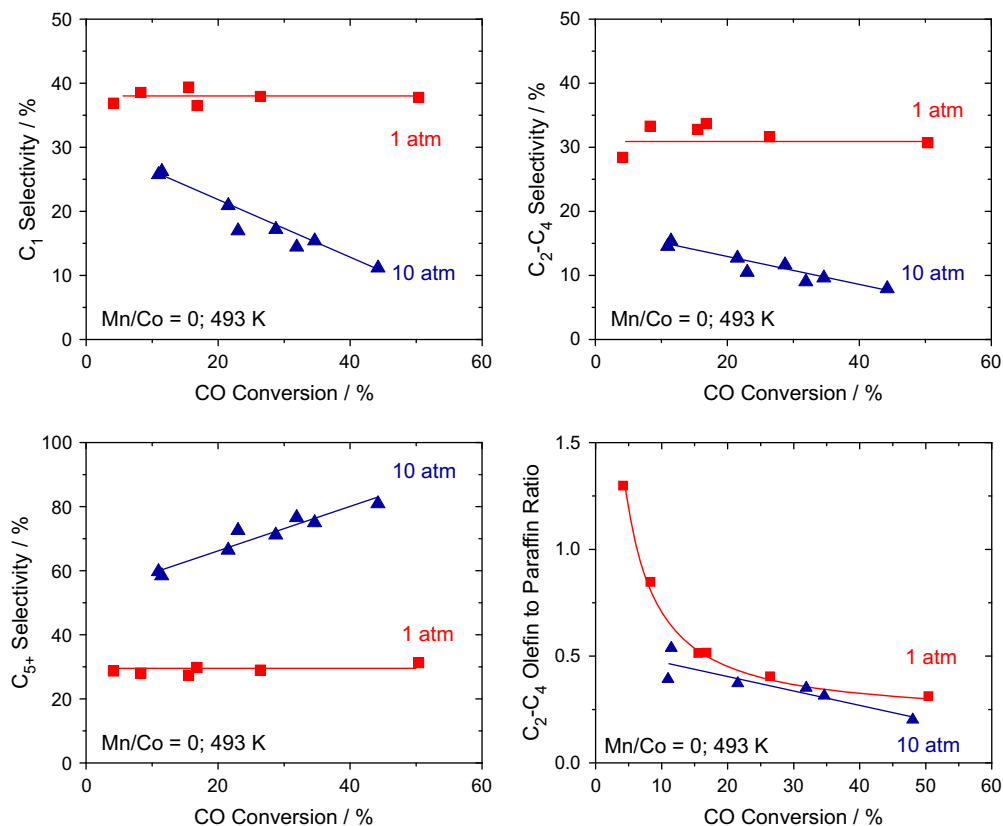


Fig. 6. Effect of pressure (1 and 10 atm) on C₁, C₂–C₄, and C₅+ selectivities and O/P ratios within the C₂–C₄ fraction at varying degrees of CO conversion for unpromoted Co/SiO₂. Temperature = 493 K, feed ratio H₂/CO/Ar = 2/1/0.2, flows = 40–120 ml/min, catalyst mass = 150 mg.

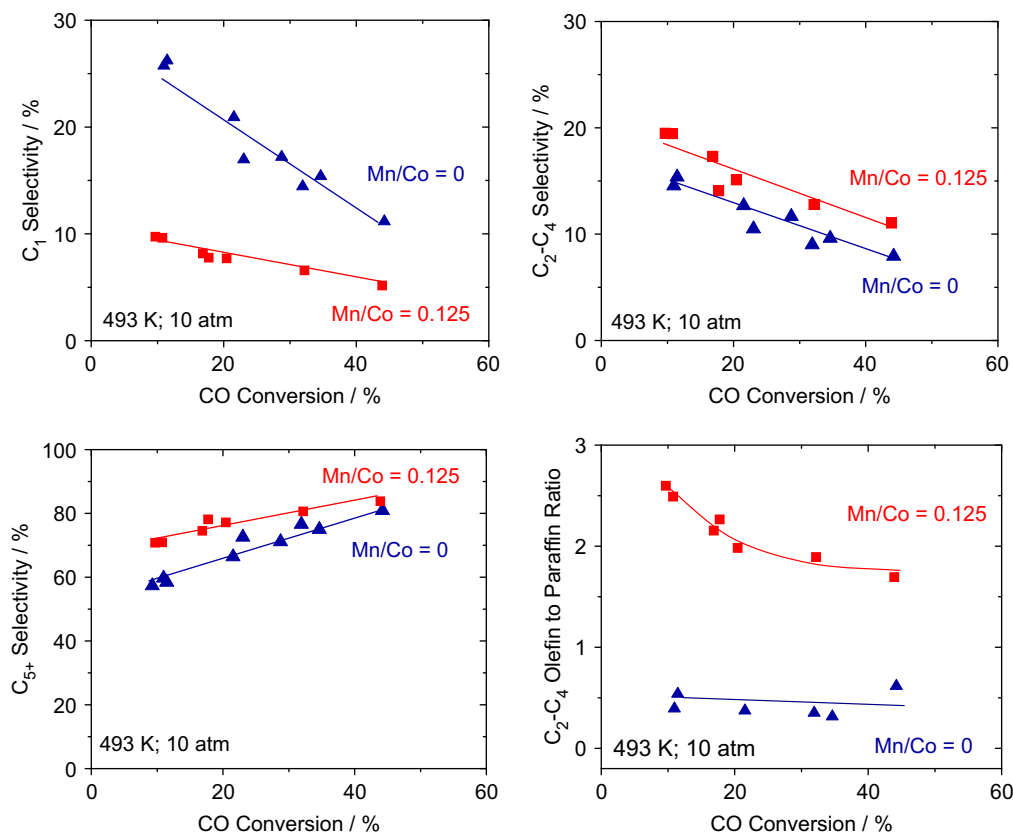


Fig. 7. Effect of Mn promotion on C_1 , C_2 – C_4 , and C_{5+} selectivities and olefin to paraffin ratios within the C_2 – C_4 fraction at high pressure (10 atm) and varying degrees of CO conversion for unpromoted Co/SiO₂ and catalysts with Mn/Co = 0.125. Temperature = 493 K, feed ratio H₂/CO/Ar = 2/1/0.2, flows = 40–120 ml/min, catalyst mass = 100 mg.

Table 4
Comparison of CO consumption and HC formation rates for unpromoted Co/SiO₂ and Mn-promoted Co/SiO₂ with a Mn/Co ratio of 0.125. Pressure = 10 atm, temperature = 493 K, conversion = 10–43%. Calculations based on values given Fig. 7.

Catalyst	CO (10 ^{−5} mol/g _{Co} /s)	Ci (10 ^{−5} mol/g _{Co} /s)	C ₂ –C ₄ (10 ^{−5} mol/g _{Co} /s)	C ₅₊ (10 ^{−5} mol/g _{Co} /s)
Mn/Co = 0	8.6–11.5	2.2–1.5	1.3–1	5.2–8.9
Mn/Co = 0.125	5.6–7.6	0.9–0.6	1.7–1.5	6–9.4

The rates of CO consumption and hydrocarbon formation with respect to C_1 , C_2 – C_4 , and C_{5+} at 10 atm for unpromoted and Mn-promoted Co/SiO₂ are summarized in Table 4 for CO conversions of 10% and 44% CO. At both levels of CO conversion, C_1 productivity was lower upon Mn promotion, and C_2 – C_4 as well as C_{5+} productivities were higher.

C_2 – C_4 olefin and paraffin selectivities were also analyzed individually as illustrated in Fig. 8. It can be seen that especially at low CO conversion, olefin selectivities were higher for Mn-promoted Co/SiO₂ and paraffin selectivities were lower. Both olefin and paraffin selectivities decreased with increasing CO conversion. This behavior was also observed for each individual product in this class (see S4 and S5).

4. Discussion

4.1. Effect of Mn promotion on catalyst structure

The results presented in Table 1 show that Mn promotion has several effects on the catalyst structure. Mn promotion decreased the average particle size of Co slightly from about 6–4 nm as the Mn/Co ratio increased from 0 to 0.5. Over this same range of Mn/Co ratios, the reducibility of Co decreased from 83% to 66%, with

most of the change occurring for the smallest Mn/Co ratio, 0.05. An approximately twofold decrease in H₂ uptake also occurred when the Mn/Co ratio was raised from 0 to 0.05. Further increasing the Mn/Co ratio up to 0.25 did not alter the H₂ uptake significantly, but increasing the Mn/Co ratio to 0.5 decreased the H₂ uptake by an order of magnitude.

An understanding of how Mn affects the structure of Mn-promoted Co/SiO₂ can be deduced from the literature. The phase diagram of the Co–Mn–O system shows that at the preparation conditions used in our study, formation of CoMnO₃, as well as a γ -Mn₃O₄ spinel and α -Mn₂O₃ in different ratios is favorable at Mn/Co ratios of higher than 0.1, while at lower ratios, formation of the spinel only can be expected [47]. It is noted, though, that none of these phases were observed in the XRD patterns of the fully oxidized catalyst, which only showed evidence for Co₃O₄. Similar findings have been reported in studies of Mn-promoted Co/TiO₂ [41,48]. The absence of any peaks for MnO_x suggests that these species are either amorphous or are present in the form of very small nanoparticles. Characterization of Mn-promoted Co/TiO₂ after reduction showed the presence of Co(0) in the vicinity of CoO, MnO, and Co_{3–x}Mn_xO₄. No evidence for metallic Mn(0) was found after reduction, and the average oxidation state of Mn was +2 [48].

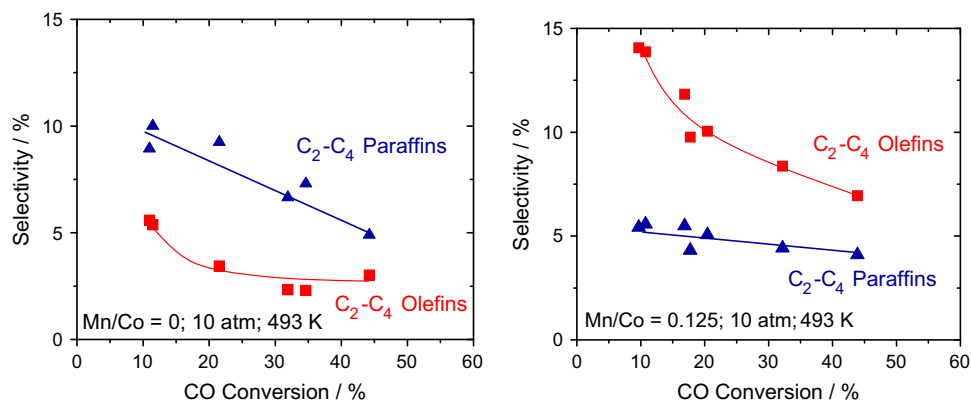


Fig. 8. Effect of Mn promotion on olefin and paraffin selectivities within the C₂–C₄ fraction at high pressure (10 atm) and varying degrees of CO conversion for unpromoted (left) and Mn-promoted (right) Co/SiO₂ with Mn/Co ratios of 0.125. Temperature = 493 K, feed ratio H₂/CO/Ar = 2/1/0.2, flows = 40–120 ml/min, catalyst mass = 100–150 mg.

Taken together, our results and those reported in the literature suggest that after reduction, 80–70% of the cobalt was present as metallic Co and that the balance was either CoO_x or Co_{3–x}Mn_xO₄. The majority of the Mn was present as MnO distributed either on the support or on the surface of the Co. The extent to which some of the Mn may enter into the surface of the Co particles to form a bimetallic is not known.

4.2. Effects of pressure and Mn promotion on activity

At 1 atm, the rates of CO consumption were well within the range of those reported previously for Co/SiO₂ [9,29] and Mn-promoted Co/Pt/SiO₂ [31] and Co/TiO₂ [39,42]. However, at 10 atm and 493 K, our rates were somewhat higher than those reported in studies using Co/SiO₂ carried out at 20 atm and 473 K [3,7,10]. The observed differences in the rate of CO consumption are most likely due to the differences in reaction conditions. Few studies report rates for Mn-promoted Co-based catalysts at high pressures [40,44]. Morales et al. found the same reaction rates at 1–18 atm for unpromoted and Mn-promoted Co/SiO₂ at 1% CO conversion. We believe that at such low CO conversion, no reliable conclusions can be drawn liquid product collection can take up to several weeks. Das and coworkers report higher reaction rates for Mn-promoted Co/SiO₂ at 21 atm which also disagrees with the data reported here.

The effects of pressure on the rate of CO conversion can be understood in terms of the mechanism for CO hydrogenation on Co proposed by Ojeda et al. shown in Scheme 1 [49]. The authors present theoretical evidence to suggest that the cleavage of the C–O bond of adsorbed CO is H-assisted. The rate law derived from the proposed mechanism is presented in Eq. (6).

$$-r_{\text{CO}} = \frac{K_1 K_2 K_6 k_7 P_{\text{H}_2} P_{\text{CO}}}{(1 + K_1 P_{\text{CO}})^2} \quad (6)$$

Here, K_i is the adsorption constant for reaction i ($i = 1, 2, 6$), k_7 is the rate coefficient for reaction 7, and P_i is the partial pressure of component i . The inhibition of CO consumption due to H₂ adsorption is neglected due to the weakness of H₂ adsorption on Co, especially in the presence of co-adsorbed CO [49,50]. The form of Eq. (6) is identical to that reported by Yates and Satterfield and was found to best represent experimental data for supported Co catalysts over a wide range of operating conditions [51]. The authors express the product of K_1 , K_2 , K_6 , and k_7 as coefficient a and K_1 as coefficient b .

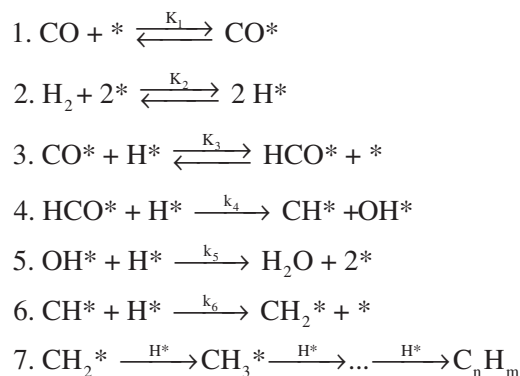
Values for the coefficients a and b were determined in the following manner. A differential mass balance was written on the reactor, assuming that the reactor behaves as a plug flow reactor, and then integrated from the reactor inlet to the reactor outlet in

order to determine the conversion of CO for a specified inlet feed composition and flow rate. The resulting expression was used to determine the integral rate of reaction, R_{int} , as a function of CO conversion and the feed partial pressures of CO and H₂, $P_{\text{H}_2}^0$ and P_{CO}^0 , respectively, Eq. (7).

$$-R_{\text{int}} = \frac{X_{\text{CO}} a P_{\text{H}_2}^0}{b(2 \ln(1 - X_{\text{CO}}) - b P_{\text{CO}}^0 X_{\text{CO}})} \quad (7)$$

The values of a and b were then adjusted to fit Eq. (7) to the observed rates of reaction measured as a function of CO conversion at both 1 atm and 10 atm.

The value of a determined in this manner is $9.6 \pm 4.4 \times 10^{-5}$ mol/g_{Co} atm² s for unpromoted Co/SiO₂, which agrees well with the value reported by Yates and Satterfield (7.5×10^{-5} mol/g_{Co} atm² s) [51]. For Mn-promoted Co/SiO₂ (Mn/Co = 0.125), the value of a is $5.1 \pm 3.6 \times 10^{-4}$ mol/g_{Co} atm² s. The differences in the values of a for Mn-promoted and unpromoted Co/SiO₂ can be attributed to two factors. The first is the difference in the concentration of reduced Co sites present on the two catalysts, and the second is the explicit effect of the promoter. Reference to Table 1 shows that Mn-promoted Co/SiO₂ exhibited a lower uptake of H₂, from which it is deduced that the concentration of reduced Co sites present on the Mn-promoted catalyst was lower. Therefore, the 7.5-fold higher value of a for Mn-promoted Co/SiO₂ reflects a rate enhancing effect of the promoter that offsets a loss in accessible Co sites. A possible explanation for this effect is that a part of the Co particle is covered by MnO_x species. As suggested by Boffa and coworkers, the Mnⁿ⁺ exposed at the edge of such species can act as Lewis-acids that enhance the adsorption of



Scheme 1. Suggested elementary steps for FTS on Co-based catalysts based on Ref. [49].

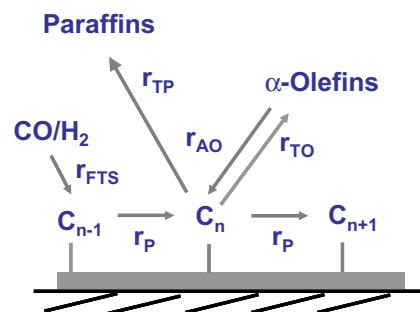
CO and its ease of hydrogenation [52]. It should be noted though this effect is sensitive to the fractional coverage of the metal by the promoting oxide and reaches a maximum for a coverage of ~ 0.05 . The loss in activity observed in the present study for bulk Mn/Co ratios above 0.05 may be due to this effect as well as to the reduction in the average size of the Co particles with increasing Mn/Co ratio as noted in Table 1. Recent work by de Jong et al. [53] has shown that the turnover frequency for FTS on carbon-supported Co increases with particle size up to a size of 4.2 nm, but is independent of particle size for larger particles. By contrast, Prieto et al. have found that the turnover frequency for FTS is independent of Co particle size only for particle sizes larger than 10 nm. [54] The results presented in Table 1 show that the average size of the particles in our experiments was greater than 4.4 nm, with the exception of the sample for which the bulk Mn/Co ratio is 0.5, in which case the particle size decreases to 3.8 nm and that all particle sizes are smaller than 10 nm. Thus, it is possible that the decrease in particle size that is observed with increasing Mn/Co ratio (see Table 1) also contributes to the loss in activity with increasing Mn/Co ratio reported in Fig. 1. The value of b in Eq. (6) was found to be $1 \pm 0.34 \text{ atm}^{-1}$ for unpromoted Co/SiO₂ and is comparable to that reported by Yates and Satterfield (2.23 atm^{-1}) at 493 K [51]. The value of b for Mn-promoted Co/SiO₂ is $3.0 \pm 1 \text{ atm}^{-1}$, suggesting that Mn promotion enhances the strength of CO adsorption.

Using the coefficients a and b determined above predicts that the rate of CO consumption should decrease with increasing CO conversion at 1 atm, in good agreement with what was observed experimentally (Fig. 1). At 10 atm, CO consumption rates should also decrease with increasing CO conversion according to Eq. (7), but our data showed no change with CO conversion. We believe that this is because Eq. (7) does not accurately reflect the activating effect of water that has been reported for Co/SiO₂ catalysts at high pressures [55,56]. At 10 atm, the higher value of b for Mn-promoted Co/SiO₂ relative to unpromoted Co/SiO₂ results in a strong inhibitory effect of CO adsorption on the former catalyst, which offsets its higher intrinsic activity as reflected in the higher value of a . The net result is that Mn-promoted Co/SiO₂ exhibits a lower rate of CO consumption at 10 atm than unpromoted Co/SiO₂, which is opposite to the relationship in activities for these two catalysts at 1 atm (see Figs. 1 and 5).

4.3. Effects of reaction conditions and Mn promotion on product distribution

The results of this investigation show that the carbon number distribution of products formed by FTS depend on the reaction pressure and the promotion of Co/SiO₂ with Mn. While the extent of CO conversion has little to no effect on product distribution at 1 atm, the effect is significant at 10 atm. The data presented in Figs. 2 and 3 show that for FTS carried out at 1 atm, Mn promotion of Co/SiO₂ causes a decrease in the C₁ selectivity, no change in the C₂–C₄ selectivity, and an increase in the C₅₊ selectivity. However, the product selectivities for both catalysts were unaffected by CO conversion. Raising the reaction pressure to 10 atm decreases the C₁ and C₂–C₄ selectivities and increased the C₅₊ selectivity for both unpromoted and promoted Co/SiO₂. Mn promotion decreases the C₁ selectivity and increases the C₂–C₄ and C₅₊ selectivities.

Reaction pressure, Mn promotion, and CO conversion also affected the O/P ratio of the C₂–C₄ products. At 1 atm, Fig. 4 shows that the C₂–C₄ fraction was almost exclusively olefinic at very low CO conversions, suggesting that olefins are the primary product of FTS synthesis on both unpromoted and Mn-promoted Co/SiO₂. With increasing CO conversion, the fraction of C₂–C₄ paraffins increased and the fraction of C₂–C₄ olefins decreased, the extent to which this change occurred was more dramatic for



Scheme 2. Suggested reaction network of FTS assuming that Co catalysts are not active for isomerization and hydroformylation of olefins at the reaction conditions used here [3]. Based on results reported in [57], it can be assumed that chain propagation is considerably faster than de-propagation.

unpromoted Co/SiO₂. Consistent with these trends, the O/P ratio of the products decreased with CO conversion, but at any CO conversion, the O/P ratio was higher for the Mn-promoted catalyst (see Fig. 3). When FTS was carried out at 10 atm, the effect of CO conversion on the distribution of C₂–C₄ olefins and paraffins was quite different from that observed at 1 atm. At the higher pressure, the selectivities to C₂–C₄ olefins and paraffins both decreased with increasing CO conversion. However, Mn promotion resulted in a much larger O/P ratio than that observed for unpromoted Co/SiO₂.

The effects of reaction pressure, CO conversion, and Mn promotion can be interpreted in the terms of the reactions shown in Scheme 2. This scheme envisions that higher molecular weight products are formed by addition of C₁ adsorbed species, e.g., CH_{2,s}, to C_n adsorbed species, e.g., C_nH_{2n+1,s}, to form C_{n+1} species [2,3]. At any stage in the chain growth process, stable products can be formed by β -hydride elimination from a growing chain to produce an α -olefin or α -hydrogen addition to produce a linear paraffin. The extent to which olefin or paraffins are formed via these primary processes depends on the intrinsic rate parameters for chain termination to olefins and paraffins, k_{to} and k_{tp} , respectively, and the coverage of the Co surface by vacancies, θ_v , needed to accommodate H atoms released by β -hydride elimination, or adsorbed H atoms, θ_H , needed for α -hydrogen addition to form paraffins. The α -olefins, the primary isomer formed during FTS, can undergo isomerization to form β - and γ -olefins, hydrogenation to form paraffins, and reactions with C₁ and C_n species to form higher molecular weight chains [3,58,18]. Secondary hydrogenation of olefins to paraffins and isomerization of α -C₄₊ olefin are observed primarily at low reaction pressures [58,18]. Both processes start by adsorption of an α -C₄₊ olefin adjacent to an H atom, resulting in the formation of either a primary or secondary alkyl species. If the secondary alkyl species undergoes β -hydride elimination, then a β -olefin will form. The isomerization of β -olefins to γ -olefins can occur via a similar process. However, if a second atom of H reacts with either the primary or secondary alkyl species, then normal paraffin is formed. At the high olefin concentrations encountered when the reaction pressure is high and at moderate to high CO conversions, α -olefin reincorporation into chain growth becomes increasingly important, particularly for C₂–C₄ olefins, while other isomers are only primary products and are not incorporated into growing chains [3].

The absence of any change in product carbon number distribution with CO conversion observed at 1 atm for both unpromoted and Mn-promoted Co/SiO₂ suggests that at this pressure, the reincorporation of olefins into chain growth is negligible and that the formation of isomers within a carbon number fraction does not affect the total carbon number distribution. The consumption ratio of H₂ to CO is nearly identical to that of the feed ratio. The latter assumption is consistent with the overall reaction stoichiometry

of $\text{CO} + 2\text{H}_2 \rightarrow -\text{CH}_2- + \text{H}_2\text{O}$. The observed changes in the O/P ratio of the C_2 – C_4 fraction with increasing CO conversion indicate that olefins once formed undergo secondary hydrogenation to form paraffins. This conclusion is supported by the data shown in Figs. 3 and 4, as well as the data for individual C_2 – C_4 products shown in Figures S2 and S3, and is identical to that reached by Kuipers and coworkers based on studies of FTS on unpromoted Co/SiO_2 [59]. We also note that at 1 atm, 1-butene undergoes isomerization to cis- and trans-2-butene, but all butene isomers decrease in concentration with increasing CO conversion (see Table S1).

Mn promotion decreases the C_1 and C_2 – C_4 selectivities and the extent of secondary hydrogenation of C_2 – C_4 olefins and increases C_{5+} selectivities. This effect is similar to that reported earlier by Weckhuysen and coworkers [39,42], who hypothesized that these effects were due to a decrease in the availability of adsorbed hydrogen on the catalyst surface. As noted earlier, this conclusion is supported by the higher value of the coefficient b in Eqs. (6) and (7) for Mn-promoted Co/SiO_2 . It is noted further that Mn promotion enhances the intrinsic O/P ratio seen at CO conversions below 5%, further supporting the conclusion that the interaction of Mn or MnO_x with Co reduces the availability of adsorbed H_2 . The reduction in the intrinsic rate of chain termination by hydrogenation of C_n species (see Scheme 2) causes the probability for chain growth to increase, thereby contributing to the increase in higher molecular weight products seen for Mn-promoted Co/SiO_2 . The data shown in Table S1 suggest that Mn promotion has only a modest effect in reducing the extent of 1-butene isomerization to 2-butenes.

The strong shift in the product distribution to higher molecular weight products observed for unpromoted Co/SiO_2 upon raising the reaction pressure from 1 atm to 10 atm was similar to that reported in previous studies [7,9–12,15]. Since this effect was also observed at low CO conversion (10%) this cannot be attributed to the effect of olefin re-incorporation alone. Thus this effect is attributed to a reduction in coverage of the catalyst surface by H atoms relative to CO. The occurrence of such an effect can be demonstrated assuming that the H_2 and CO adsorption are equilibrated. In this case, the coverage of H and CO, θ_{H} , and θ_{CO} are given by Eqs. (8) and (9), respectively.

$$\theta_{\text{H}} = K_{\text{H}_2}^{1/2} P_{\text{H}_2}^{1/2} \theta_{\text{V}} \quad (8)$$

$$\theta_{\text{CO}} = K_{\text{CO}} P_{\text{CO}} \theta_{\text{V}} \quad (9)$$

Here, K_{H_2} and K_{CO} represent the equilibrium constants for H_2 and CO adsorption, respectively, and P_{H_2} and P_{CO} are the partial pressures of H_2 and CO, respectively. Recognizing that $P_{\text{H}_2} = 2/3P(1 - X_{\text{CO}}^{\text{CO}})$ and $P_{\text{CO}} = 1/3P(1 - X_{\text{CO}}^{\text{CO}})$, where P is the total pressure and $X_{\text{CO}}^{\text{CO}}$ is the conversion of CO, leads to the result that

$$\frac{\theta_{\text{H}}}{\theta_{\text{CO}}} = \sqrt{6} \frac{K_{\text{H}_2}^{1/2}}{K_{\text{CO}}} \frac{1}{P^{1/2}(1 - X_{\text{CO}}^{\text{CO}})^{1/2}} \quad (10)$$

From Eq. (10), it is evident that as P increases, the ratio $\theta_{\text{H}}/\theta_{\text{CO}}$ decreases. A decrease in this ratio causes chain termination by hydrogenation of C_n species to decrease. Increasing the total pressure also results in a decrease in the vacancy fraction θ_{V} , which in turn leads to a reduction in the rate of chain termination by β -hydride elimination. Taken together, these effects reduce the rate of chain termination relative to chain propagation, causing the molecular weight distribution of the products to shift toward higher carbon numbers. Mn promotion further enhances these effects by reducing the availability of adsorbed H and strengthening the adsorption of CO, i.e., increasing the value of K_{CO} .

Two other factors may contribute to the decrease in C_1 and the increase in C_{5+} selectivities observed at high pressure and CO conversion. Under these conditions, the partial pressure of water is

high, and prior studies have shown that water vapor can alter the product distribution in the direction noted the present work [56]. The condensation of C_{5+} olefins within the catalyst pores can also contribute to olefin reincorporation and hence to a reduction in the formation of methane and an increase in the formation of higher molecular weight products [60]. However, as seen in Table 4, the increase in the formation rate of C_{5+} products upon Mn promotion at high pressures and CO conversions is $\sim 6\%$ and, therefore, the effects of liquid accumulation in the catalyst pores are expected to make only a minor contribution to altering the product distribution.

As noted earlier, at 10 atm, the distribution of products shifted to higher molecular weights with increasing CO conversion, in strong contrast with what was observed at 1 atm. Another difference was that at 10 atm, the selectivities to C_2 – C_4 paraffins, as well as olefins, decreased with increasing CO conversion. Similar findings have been reported by Iglesia et al. [3]. These trends are attributable to reincorporation of C_2 – C_4 olefins into the chain growth process. As the CO conversion increases, the partial pressures of the C_2 – C_4 olefins increased, further enhancing their incorporation and resulting in a shift in the carbon number distribution of the products to higher molecular weights. Mn promotion increases the selectivity to C_2 – C_4 products and the O/P ratio of these products, thereby providing even greater partial pressures of C_2 – C_4 olefins and contributing further to the reincorporation of these products into chain growth. Thus, the shift to higher molecular weight products at high pressure can be attributed to a reduction in the ratio of H to CO adsorbed on the catalyst and to the reincorporation of C_2 – C_4 olefins. The influence of olefin reincorporation becomes more pronounced as the concentration of these products increases with increasing residence time (i.e., increasing CO conversion). The promotion of Co/SiO_2 with Mn enhances both of these effects, resulting in very low yields of methane ($< 5\%$ at 50% CO conversion) and high yields of C_{5+} products ($> 85\%$ at 50% CO conversion). Mn promotion also has the effect of strongly suppressing the isomerization of 1-butene to 2-butene (see Table S2). This change is also attributed to the effect of Mn reducing the availability of H atoms on the surface of the catalyst.

5. Conclusions

The following conclusions are drawn from the present study:

- (1) At 1 atm, Mn promotion of SiO_2 -supported Co caused the rate of CO conversion to pass through a maximum at $\text{Mn/Co} = 0.05$. This pattern is believed to be due to the interplay between the two effects of MnO deposited onto the surface of the supported Co particles. The first is an increase in the activity of Co sites located adjacent to MnO as a consequence of the promotion of CO dissociation resulting from the Lewis acid–base interaction of Mn^{2+} cations with the O atom of CO bound to Co through the C atom. The second is a decrease in the number of exposed Co sites, which occurs as the Mn/Co ratio increases. The rate of CO consumption decreased with CO conversion, consistent with the rate expression given by Eq. (6), which predicts that at 1 atm, the rate of reaction should be proportional to the product of H_2 and CO partial pressures.
- (2) At 10 atm, Mn promotion ($\text{Mn/Co} = 0.125$) caused a decrease in the rate of CO consumption. This effect is attributed to the enhanced equilibrium constant for CO adsorption, which results in inhibition of CO hydrogenation. While the rate expression for CO consumption, Eq. (6), correctly predicts a decreasing reaction rate with increasing CO conversion at 1 atm, it does not account for the observed conversion-independent rate for unpromoted catalyst or the slight increase

in the rate with conversion for Mn-promoted catalyst. It is suggested that this effect may be attributable to the ability of water vapor at high partial pressures to enhance the rate of CO hydrogenation.

- (3) Increasing total pressure from 1 atm to 10 atm led to a strong decrease in the formation of methane and an increase in the formation of C_5+ products, but had little effect on the selectivity to C_2 – C_4 products. Similar effects were observed for both unpromoted and Mn-promoted Co/SiO₂; however, the effects were more pronounced for the promoted catalyst. It was also observed that while C_2 – C_4 olefins underwent secondary hydrogenation to paraffins at 1 atm, this process was almost completely suppressed at 10 atm and replaced by olefin reincorporation into hydrocarbon chain growth. This can be explained based on first principles that predict that H/CO ratios on the catalyst surface decrease with increasing total pressure. At 10 atm, the chain incorporation of olefins led to a decrease in C_1 – C_4 selectivities with increasing CO conversion while C_5+ selectivities increased.
- (4) Mn promotion led to an increase in the fraction of C_5+ products formed at all operating conditions. The O/P ratio of the C_2 – C_4 fraction was higher upon Mn promotion as a consequence of an increase in the primary formation of olefins and a decrease in the primary formation of paraffins. At 1 atm, Mn promotion also suppressed secondary hydrogenation of olefins to paraffins. These effects can be attributed to an increase in the equilibrium constant of CO adsorption relative to that for H₂ adsorption, which results in a reduction in the H/CO ratio on the catalyst surface.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.jcat.2012.01.008.

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